

Wenkai Li

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EDUCATION

Xi'an Jiaotong University, China

Advisor: Prof. Peng Zhang

Research focus: Network Simulation, Network Verification, Formal Analysis for Networked System

M.S. in Computer Science and Technology

Sep. 2024 – Present

Lanzhou University, China

Advisor: Prof. Xin Liu

B.S. in Computer Science and Technology

Sep. 2020 – Jun. 2024

PUBLICATIONS

1. **Wenkai Li**, Ran Shu, Peng Zhang, Yiren Zhao, Danfeng Shan, Yongqiang Xiong
Nüwa: A Generative Control Plane for AI Network Simulation
To appear in Proceedings of the ACM SIGCOMM 2026 Conference (SIGCOMM '26).
2. **Wenkai Li**, Ran Shu, Peng Zhang, Yongqiang Xiong
Nüwa: Efficient Generative Control Plane for AI Network Simulation
The 9th Asia-Pacific Workshop on Networking (APNet '25).
3. Dan Wang, Peng Zhang, Wenbing Sun, **Wenkai Li**, Xing Feng, Hao Li, Jiawei Chen, Weirong Jiang, Yongping Tang
S2: A Distributed Configuration Verifier for Hyper-Scale Networks
Proceedings of the ACM SIGCOMM 2025 Conference (SIGCOMM '25).

AWARDS & SCHOLARSHIPS

Excellent Graduate Student, Xi'an Jiaotong University	2025
<i>Awarded in recognition of outstanding overall academic and extracurricular performance</i>	
Huawei Scholarship	2025
<i>Awarded to 5 of 1,000+ students for excellence in academics, research, and competitions</i>	
Academic Scholarship, Xi'an Jiaotong University	2025
<i>Awarded in recognition of outstanding academic performance</i>	
APNet Student Grant	2025
<i>Awarded to student attendees selected to receive travel support for APNet 2025</i>	
Gold Medal in ICPC Asia Xi'an Regional Contest	2023
<i>Ranked 26th among 379 teams</i>	
Gold Medal in ICPC Asia Shenyang Regional Contest	2022
<i>Ranked 17th among 734 teams</i>	
Silver Medal in ICPC Asia East Continent Final Contest	2022
<i>Ranked 64th among 360 teams</i>	
Silver Medal in ICPC Asia Xi'an Regional Contest	2022
<i>Ranked 98th among 516 teams</i>	
Silver Medal in China Collegiate Programming Contest (CCPC) Guilin Site	2022
<i>Ranked 67th among 322 teams</i>	
Huawei Intelligent Base Scholarship	2022
<i>Awarded in recognition of outstanding performance in academics and innovation</i>	
Bronze Medal in China Collegiate Programming Contest (CCPC) Guangzhou Site	2021
<i>Ranked 111th among 245 teams</i>	
Academic Scholarship, Lanzhou University	2021 - 2023
<i>Awarded in recognition of outstanding academic performance</i>	
Silver Medal in ICPC Asia Shanghai Regional Contest	2021
<i>Ranked 41th among 245 teams</i>	
Second Prize, 24th National Olympiad in Informatics in Provinces (NOIP)	2018
<i>Ranked 30% among 12856 competitors</i>	

RESEARCH PROJECTS

Network Verification

Formal Methods for Shared-Buffer Network Performance Analysis Nov. 2025 – Present

- Built a packet-level logical model for shared-buffer datacenter switches.
- Encoded network performance metrics as logical formulas to support formal performance analysis.
- Analyzed buffer management policies using the proposed logical model.

S2: A Distributed Configuration Verifier for Hyper-Scale Networks Nov. 2023 – Jan. 2025

- Implemented a distributed control-plane simulation engine, offloading route exchange with nodes on remote hosts.
- Designed a distributed BDD data structure for verifying data-plane properties based on distributed routing results.
- Verifying a real 16K-switch DCN with 200M IPv4 routes in 16 minutes using 35GB memory, and scaling to 10K+ switches and 1000M routes with 16 workers.

Network Simulation

Nüwa: Efficient Generative Control Plane for AI Network Simulation Nov. 2024 – Feb. 2026

- Developed an efficient control-plane routing construction framework for large-scale AI network simulation.
- Formulated routing as a compilation problem, compiling topology and routing policies into compact forwarding artifacts.
- Reduced initialization time from hours to 25 seconds for a 65,536-GPU cluster, cut end-to-end simulation memory footprint to 20% of existing approaches on a 40K+ GPU cluster, and scaled to 221,184 GPUs.

INDUSTRY EXPERIENCE

Microsoft Research Asia | Research Intern | Network Infrastructure Group Beijing, China

Efficient Simulator for Large-Scale AI Network Nov. 2024 – Sep. 2025

- Implemented Nüwa on top of ns-3.
- Built a compilation pipeline that converts declarative topology and routing inputs into compact forwarding artifacts.
- Developed structured forwarding and prefix-scoped exception mechanisms for shortest-path routing, policy-based non-minimal routing, and link/switch failure handling.

TEACHING EXPERIENCE

Xi'an Jiaotong University 2025

Teaching Assistant - Software Defined Networks, COMP550705

Shanxi Xiangyang Competitive Programming Training Institute 2024

Coach - Algorithm and Data Structure

Lanzhou University 2021

Teaching Assistant - Summer School, Computer Programming

SERVICE & TALK

Artifact Evaluation Committee

- MLSys 2026
- CAIS 2026

Nüwa: Efficient Generative Control Plane for AI Network Simulation

- Talk at APNet 2025, Shanghai, China

TECHNICAL SKILLS

Programming Languages: C++/C, Java, Python, JavaScript, LATEX

Systems & Tools: Docker, Git, Linux

Development: SQLite, PostgreSQL, React, Node.js, VSCode